

# Detecting Poisoned Frames in Multi-Camera Tracking

10-Week Schedule

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|                                          |                                           |                                                  |                                                   |
|------------------------------------------|-------------------------------------------|--------------------------------------------------|---------------------------------------------------|
| <b>Phase 1 — Foundation</b><br>Weeks 1–2 | <b>Phase 2 — Exploration</b><br>Weeks 3–4 | <b>Phase 3 — Assess &amp; Build</b><br>Weeks 5–7 | <b>Phase 4 — Writing &amp; Pub.</b><br>Weeks 8–10 |
|------------------------------------------|-------------------------------------------|--------------------------------------------------|---------------------------------------------------|

**Seed Papers: 4 Assigned, Weeks 1–2.**  
**Students will find all other sources independently.**

| # | Week / Who       | Title & Authors                                | Venue                       | Link                                                                                            |
|---|------------------|------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------|
| 1 | Wk 1 · Christina | <b>Wild Patterns Reloaded — Cinà et al.</b>    | <i>ACM Surveys</i> 2023     | <a href="https://arxiv.org/abs/2205.01992">arxiv.org/abs/2205.01992</a>                         |
| 2 | Wk 1 · Floyd     | <b>ByteTrack — Zhang et al.</b>                | <i>ECCV</i> 2022            | <a href="https://arxiv.org/abs/2110.06864">arxiv.org/abs/2110.06864</a>                         |
| 3 | Wk 2 · Both      | <b>BadNets — Gu et al.</b>                     | <i>IEEE Access</i> 2019     | <a href="https://arxiv.org/abs/1708.06733">arxiv.org/abs/1708.06733</a>                         |
| 4 | Wk 2 · Both      | <b>Blinding &amp; Blurring the MOT Tracker</b> | <i>Neural Networks</i> 2024 | <a href="https://doi.org/10.1016/j.neunet.2024.106248">doi.org/10.1016/j.neunet.2024.106248</a> |

# PHASE 1: FOUNDATION

Weeks 1 - 2

## Week 1

PHASE 1  
FOUNDATION

### Orientation & Research Landscape

*Project stack introduced, seed papers assigned, environment setup begins*

#### LEARNING OBJECTIVES

- Understand the PRIME mission and where this project sits within ML security research
- Learn the three-layer architecture of the revised project: BoT-SORT-ReID + CityFlowV2 + embedding-space attack
- Understand what a ReID embedding is and why it is the correct attack surface for multi-camera tracking
- Learn how to read, annotate, and present a research paper using the 6-point format

#### WEEKLY GOALS

- Both students can explain in plain language what a ReID embedding does in a multi-camera tracker
- Research environment partially set up: Python env created, BoT-SORT repo cloned
- CityFlowV2 Scenario S01 downloaded directly from aicitychallenge.org (no registration required)
- First Friday paper presentation delivered using the 6-point format

#### STUDENTS

- Christina: read Seed 1 — Cinà et al., Wild Patterns Reloaded (poisoning survey, 2023)
- Floyd: read Seed 2 — Zhang et al., ByteTrack ECCV 2022 (MOT pipeline foundations)
- Both: submit annotated bibliography entry for assigned paper before Friday
- Both: clone BoT-SORT repo, create conda environment, confirm Python and PyTorch install

#### RESEARCHER

- Orient students to PRIME mission, 10-week arc, and the revised project stack
- Explain why ByteTrack alone is insufficient: no ReID = no identity layer = no interesting attack surface
- Demonstrate what a ReID embedding looks like: run OSNet on two crop images, print cosine similarity
- Download CityFlowV2 Scenario S01 directly from aicitychallenge.org — no registration required

#### FRIDAY PRESENTATION

- Christina: 10-min presentation on Seed 1 — what is data poisoning, what types exist, why does it matter for vision?
- Floyd: 10-min presentation on Seed 2 — how does ByteTrack work and what are MOTA and IDF1?
- Group discussion: ByteTrack has no appearance embeddings — what does that mean for an identity-layer attack?
- Researcher bridges to next week: BoT-SORT adds ReID on top of ByteTrack — that is our attack surface

#### DELIVERABLES

- Annotated bibliography entry — 1 per student, submitted Thursday before Friday session
- BoT-SORT repo cloned and conda environment created on each student machine
- CityFlowV2 Scenario S01 downloaded and linked to BoT-SORT data directory
- End-of-week reflection note: what is a ReID embedding and why does it matter for multi-camera tracking?

**SEED** Seed 1 → Christina: [arxiv.org/abs/2205.01992](https://arxiv.org/abs/2205.01992) (Cinà et al., 2023 — poisoning survey)

**SEED** Seed 2 → Floyd: [arxiv.org/abs/2110.06864](https://arxiv.org/abs/2110.06864) (Zhang et al., 2022 — ByteTrack)

## Week 2

PHASE 1  
FOUNDATION

### Threat Modeling & The ReID Attack Surface

*BadNets mechanics applied to embedding space; multi-camera attack surface mapped*

#### LEARNING OBJECTIVES

- Understand how BadNets backdoor mechanics translate from image classification to ReID embedding poisoning

#### WEEKLY GOALS

- Each student produces a threat model diagram showing all attack entry points in a BoT-SORT-ReID pipeline

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| <ul style="list-style-type: none"> <li>• Map the full attack surface of a multi-camera tracking system with a ReID-based association module</li> <li>• Understand what cross-camera association means: how BoT-SORT-ReID links a track from camera A to camera B</li> <li>• Begin building independent literature search skills using arXiv, Google Scholar, IEEE Xplore</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>• Both students can explain the difference between pixel-level attacks (ineffective) and embedding-space attacks (the real threat)</li> <li>• At least 1 independently found paper each — not assigned by the researcher</li> <li>• BoT-SORT-ReID running on a test video clip with ReID enabled</li> </ul>                                                                                                                                                                                   |
| <p><b>STUDENTS</b></p> <ul style="list-style-type: none"> <li>→ Both: read Seed 3 (BadNets) and Seed 4 (adversarial attacks on MOT trackers)</li> <li>→ Group discussion: how do BadNets embedding-space mechanics apply to ReID in multi-camera tracking?</li> <li>→ Each student independently maps the BoT-SORT-ReID attack surface — where does embedding poison enter?</li> <li>→ Each student finds 1 additional paper independently; annotate using the standard template</li> </ul>                                                                                                                                                                                                                                                                                                                                                                | <p><b>RESEARCHER</b></p> <ul style="list-style-type: none"> <li>→ Walk through BoT-SORT-ReID codebase: show exactly where OSNet extracts embeddings (fast_reid module)</li> <li>→ Demonstrate why pixel noise fails: show that adding noise to an image barely shifts the cosine similarity of the extracted embedding</li> <li>→ Demonstrate why embedding shift works: manually add a small vector to an embedding and show the cosine distance jump</li> <li>→ Confirm CityFlowV2 access — if delayed, prepare MOT17 multi-sequence fallback plan</li> </ul> | <p><b>FRIDAY PRESENTATION</b></p> <ul style="list-style-type: none"> <li>→ Both students present their threat model diagrams — where do they agree and disagree on attack entry points?</li> <li>→ Each student presents their self-found paper using the 6-point format</li> <li>→ Key group discussion: at what layer of the pipeline does poison need to enter to break cross-camera identity association?</li> <li>→ Researcher closes: the attack we will implement in week 4 modifies embedding vectors, not pixels</li> </ul> |
| <p><b>DELIVERABLES</b></p> <ul style="list-style-type: none"> <li>□ Threat model diagram: 1 per student showing attack entry points in a BoT-SORT-ReID multi-camera pipeline</li> <li>□ Annotated bibliography updated to 3 sources per student (2 seeds + 1 self-found)</li> <li>□ BoT-SORT-ReID running on a test clip with --with-reid flag active and embeddings confirmed in logs</li> <li>□ Week 2 reflection: what is the difference between pixel-level and embedding-space poisoning, and why does it matter?</li> </ul> <p><b>SEED</b> Seed 3 → Both students: <a href="https://arxiv.org/abs/1708.06733">arxiv.org/abs/1708.06733</a> (Gu et al., BadNets, 2019)</p> <p><b>SEED</b> Seed 4 → Both students: <a href="https://doi.org/10.1016/j.neunet.2024.106248">doi.org/10.1016/j.neunet.2024.106248</a> (adversarial MOT attacks, 2024)</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## PHASE 2: EXPLORATION

Weeks 3 - 4

### Week 3

PHASE 2  
EXPLORATION

### Dataset Setup & Clean Baseline

*CityFlowV2 loaded, BoT-SORT-ReID baseline benchmarked on clean multi-camera data*

#### LEARNING OBJECTIVES

- Understand the CityFlowV2 dataset structure: scenarios, cameras per scenario, synchronized video, and cross-camera annotations
- Run BoT-SORT-ReID on CityFlowV2 Scenario S01 (clean data) and record baseline IDF1, HOTA, and IDS
- Understand why IDF1 is the primary metric for this project: it directly measures cross-camera identity consistency
- Confirm that the ReID embedding module is active and producing per-detection feature vectors

#### WEEKLY GOALS

- BoT-SORT-ReID running on CityFlowV2 S01 (5 cameras, clean data) with baseline IDF1 and HOTA logged
- Both students can read the metric output and explain what each number means physically
- Embedding extraction confirmed: students can print the embedding vector for a single detection and visualize the distribution
- 1–2 independently found papers each on multi-camera tracking evaluation or ReID benchmarks

#### STUDENTS

- Set up CityFlowV2 dataset in the BoT-SORT data directory following the installation guide
- Run BoT-SORT-ReID on Scenario S01 (clean, all 5 cameras active) — log IDF1, HOTA, MOTA, IDS
- Print and inspect the embedding vectors for 10 detections: confirm shape (512-d or 256-d), range, and that two crops of the same vehicle are close in cosine space
- Find 1–2 papers independently on multi-camera tracking, cross-camera ReID, or MTMC benchmarks

#### RESEARCHER

- Confirm dataset directory structure is correct and evaluation script runs without error
- Deepen literature review on detection methods — specifically: how do existing papers detect compromised camera feeds?
- Begin drafting the problem formalization: define clean embedding distribution vs. poisoned embedding distribution formally

#### FRIDAY PRESENTATION

- Each student presents 1–2 self-found papers on MTMC tracking or ReID evaluation
- Live demo: run BoT-SORT-ReID on S01, walk through metric output together, confirm baseline numbers
- Group discussion: which metric is most sensitive to cross-camera identity failures — and why is IDF1 the primary one for this project?
- Researcher frames week 4: we will shift the embedding vectors of cameras 1 and 2 in S01 and measure the damage

#### DELIVERABLES

- BoT-SORT-ReID baseline on CityFlowV2 S01 — clean run with IDF1, HOTA, MOTA, and IDS logged
- Embedding inspection report — shape, distribution, and a cosine similarity sanity check (same vehicle = high similarity)
- 1–2 annotated bibliography entries per student from independently found papers

### Week 4

PHASE 2  
EXPLORATION

### Embedding-Space Poisoning — Two-Camera Attack

*Embedding vectors from 2 of 5 cameras shifted; cross-camera identity collapse measured*

#### LEARNING OBJECTIVES

- Implement embedding-space poisoning by intercepting the ReID feature extraction step in BoT-SORT-ReID

#### WEEKLY GOALS

- Poisoning hook implemented in BoT-SORT-ReID's fast\_reid module — adding a perturbation vector to selected camera embeddings

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| <ul style="list-style-type: none"> <li>• Understand the difference between random embedding perturbation and targeted perturbation (shifting toward a specific identity cluster)</li> <li>• Quantify cross-camera identity collapse: how much does IDF1 drop when 2 of 5 cameras produce poisoned embeddings?</li> <li>• Document the experimental setup with enough detail for full reproducibility</li> </ul>                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>• IDF1, HOTA, and IDS measured on CityFlowV2 S01 with cameras 1 and 2 poisoned at three perturbation magnitudes</li> <li>• Side-by-side comparison table: clean baseline vs. poisoned at low, medium, and high perturbation</li> <li>• Both students can explain why IDS increases and IDF1 drops — physically, not just numerically</li> </ul>                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <p><b>STUDENTS</b></p> <ul style="list-style-type: none"> <li>→ Implement the embedding poison hook: in <code>fast_reid/fastreid/engine/defaults.py</code> or the feature extraction call, add a perturbation vector to embeddings from the two designated cameras</li> <li>→ Run three experiments: perturbation magnitude <math>\epsilon = 0.1, 0.5, 1.0</math> (in L2 norm) on cameras c01 and c02 of S01</li> <li>→ Build the comparison table: IDF1, HOTA, MOTA, IDS for clean + three poisoned conditions</li> <li>→ Find 1 paper on embedding-space attacks or ReID adversarial examples</li> </ul> | <p><b>RESEARCHER</b></p> <ul style="list-style-type: none"> <li>→ Guide the poisoning hook implementation — show students exactly which function call to intercept</li> <li>→ Introduce targeted vs. untargeted perturbation: random shift vs. shift toward the centroid of a different identity cluster</li> <li>→ Begin drafting the problem formalization section for the paper: define the threat model formally</li> </ul> | <p><b>FRIDAY PRESENTATION</b></p> <ul style="list-style-type: none"> <li>→ Students present poisoning results: which metric dropped the most? Was it IDF1 or IDS, and does that match the theory?</li> <li>→ Group discussion: at what perturbation magnitude does the tracker break down — and is there a threshold or is it gradual?</li> <li>→ Each student presents 1 self-found paper on embedding-space attacks or adversarial ReID</li> <li>→ Researcher frames week 5: before we build a detector, we need to understand what we have so far</li> </ul> |
| <p><b>DELIVERABLES</b></p> <ul style="list-style-type: none"> <li>□ Embedding poison hook — implemented in BoT-SORT-ReID <code>fast_reid</code> module, confirmed working</li> <li>□ Comparison table: IDF1, HOTA, MOTA, IDS for clean and three poisoned conditions (<math>\epsilon = 0.1, 0.5, 1.0</math>)</li> <li>□ Experimental setup write-up — sufficient detail for reproducibility (camera IDs, perturbation method, seed)</li> <li>□ 2+ annotated bibliography entries per student from independently found papers</li> </ul>                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## PHASE 3: ASSESSMENT & BUILD

Weeks 5 - 7

### Week 5

PHASE 3  
ASSESSMENT  
& BUILD

### Midpoint Assessment

*5-week student review — rubric evaluated, paper argument direction confirmed*

#### LEARNING OBJECTIVES

- Demonstrate command of the research question, experimental results, and literature encountered in weeks 1–4
- Articulate the paper's central claim in one sentence: what does this work show that did not exist before?
- Identify the strongest and weakest areas of each student's contribution
- Confirm or revise the paper's research direction and detection hypothesis for weeks 6–7

#### WEEKLY GOALS

- Each student delivers a 20-minute midpoint presentation covering all 4 rubric domains
- Researcher completes the formal Week 5 rubric assessment with written narrative for each student
- Annotated bibliography at 8 or more sources per student
- One-sentence paper contribution statement agreed on by the full group before end of Friday

#### STUDENTS

- Prepare 20-minute formal presentation: weeks 1–4 work, literature, experimental findings, and one detection hypothesis
- Annotated bibliography must be at 8+ sources before this session — do not arrive without it
- Articulate: if this project produces a paper, what is the one thing it demonstrates that no prior paper has shown?

#### RESEARCHER

- Complete the Week 5 rubric for each student — scores plus written narrative in every domain
- Write an updated research direction memo: given what has been learned, what is the strongest paper angle for weeks 6–10?
- Schedule a one-on-one feedback session with each Christinafter the presentations

#### FRIDAY PRESENTATION

- Midpoint presentations ARE the Friday session — 20 minutes per student, extended discussion
- Researcher gives rubric scores and verbal feedback in the session
- Group closes by agreeing on the contribution statement and the detection approach to pursue in week 6

#### DELIVERABLES

- Midpoint presentation — 20 minutes per student covering all 4 rubric domains
- Researcher rubric assessment — completed with written narrative notes for each student
- Annotated bibliography at 8 or more sources per student
- Research direction memo and agreed-upon contribution statement

**MILESTONE:** Formal 5-week assessment. Rubric scored. Detection hypothesis and paper contribution locked.

### Week 6

PHASE 3  
ASSESSMENT  
& BUILD

### Cross-Camera Embedding Consistency Detector

*Detection method implemented: cosine distance distribution analysis across camera pairs*

#### LEARNING OBJECTIVES

- Understand the detection signal: poisoned cameras produce systematically shifted cosine distance distributions vs. clean cameras

#### WEEKLY GOALS

- Cross-camera embedding consistency detector implemented and producing per-camera anomaly scores

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| <ul style="list-style-type: none"> <li>Implement the cross-camera embedding consistency check as a concrete, runnable detector</li> <li>Understand precision and recall in the context of poisoned camera identification</li> <li>Identify the detection threshold: at what distance distribution shift do we flag a camera as poisoned?</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Precision, recall, and F1 for poisoned camera identification at all three perturbation levels from week 4</li> <li>Visualization: distribution plots of cosine distances for clean vs. poisoned cameras (the signal is visible)</li> <li>Both students can explain why this detection signal works without requiring ground truth identity labels</li> </ul>                                                                       |
| <b>STUDENTS</b> <ul style="list-style-type: none"> <li>Implement the cross-camera cosine distance detector: for each pair of cameras, collect pairwise cosine distances for all detections the tracker assigns the same global ID. Compute mean and variance per camera. Flag cameras whose distribution mean is statistically shifted from the group.</li> <li>Test against week 4 poisoned scenarios: <math>\epsilon = 0.1, 0.5, 1.0</math> — compute precision and recall at each level</li> <li>Plot the cosine distance distributions: a histogram per camera clearly showing the shift in poisoned cameras</li> <li>Find 1–2 papers on anomaly detection, distribution shift detection, or statistical process control</li> </ul> | <b>RESEARCHER</b> <ul style="list-style-type: none"> <li>Survey the detection literature more deeply — identify 2–3 prior approaches to compare against</li> <li>Draft the related work section outline for the paper</li> <li>Review the detector implementation for statistical correctness before week 7 scaling tests</li> </ul> | <b>FRIDAY PRESENTATION</b> <ul style="list-style-type: none"> <li>Students present detector results: precision, recall, F1 per perturbation level, and distribution plots</li> <li>Each student presents 1–2 papers on anomaly detection or distribution shift</li> <li>Group discussion: does the detector still work when only 1 camera is poisoned? When 3 are? Where is the boundary?</li> <li>Researcher frames week 7: can this method hold up at scale?</li> </ul> |
| <b>DELIVERABLES</b> <ul style="list-style-type: none"> <li>Cross-camera embedding consistency detector — implemented and producing per-camera anomaly scores</li> <li>Precision, recall, F1 table — poisoned camera identification at <math>\epsilon = 0.1, 0.5, 1.0</math></li> <li>Cosine distance distribution plots — clean vs. poisoned cameras, clearly showing the shift</li> <li>2+ annotated bibliography entries per student from independently found papers</li> </ul>                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

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| <b>Week 7</b><br>PHASE 3<br>ASSESSMENT<br>& BUILD | <b>Sustainability &amp; Scalability Review</b><br><i>Detector tested across scenarios, perturbation types, and boundary conditions; paper argument locked</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                   | <div> <b>LEARNING OBJECTIVES</b> <ul style="list-style-type: none"> <li>Test whether the cross-camera embedding consistency detector generalizes beyond Scenario S01</li> <li>Determine the detection boundary: at what perturbation magnitude and proportion of poisoned cameras does the method fail?</li> <li>Document failure modes with specific hypotheses — not just 'it failed' but why</li> <li>Lock the paper's core argument: what exactly does this work demonstrate, and what are its honest limits?</li> </ul> </div> <div> <b>WEEKLY GOALS</b> <ul style="list-style-type: none"> <li>Detector tested on CityFlowV2 S02 and S03 (different scenarios, different camera counts and fields of view)</li> <li>Boundary condition experiments completed: vary number of poisoned cameras from 1 to N-1</li> <li>Failure mode analysis written up with specific mechanistic hypotheses</li> <li>Paper contribution statement finalized — one sentence, no hedging, agreed on by the full group</li> </ul> </div> |



### STUDENTS

- Test detector on S02 and S03 — same attack parameters as week 4, fresh data
- Boundary condition experiment: poison 1, 2, 3, 4 cameras of a 5-camera scenario — plot precision and recall vs. number of poisoned cameras
- Test with targeted perturbation (shifting embeddings toward a specific other identity cluster) vs. random shift — does the detector behave differently?
- Find 1–2 papers on robustness, generalization, or detection of adversarial examples in vision systems

### RESEARCHER

- Review scalability findings — determine whether the contribution is best framed as a method paper or a benchmark paper
- Lock the paper's core argument this week: draft the abstract for group review
- Confirm publication target list and check call-for-paper deadlines for relevant venues

### FRIDAY PRESENTATION

- Students present scalability results: where does the detector hold, where does it fail, and why?
- Each student presents papers on robustness and generalization
- Group agrees on paper contribution statement — one sentence, agreed by all three
- Researcher previews week 8: writing begins Monday

### DELIVERABLES

- Scalability report — detector performance across S01, S02, S03, and boundary condition experiments
- Failure mode analysis — documented cases where the detector fails with mechanistic hypotheses
- Paper contribution statement — one sentence, agreed on by the full group
- Draft abstract from lead researcher

★ **MILESTONE:** Scalability review complete. Paper argument locked. Writing phase begins next week.

## PHASE 4: WRITING & PUBLICATION

Weeks 8 - 10

### Week 8

PHASE 4  
WRITING &  
PUBLICATION

### Paper Draft — Literature Review & Introduction

*Seed papers anchor the lit review structure; introduction and contribution framed*

#### LEARNING OBJECTIVES

- Organize the literature review thematically across four pillars: data poisoning foundations, MOT pipeline, ReID and cross-camera tracking, adversarial attacks on trackers
- Understand how to frame a contribution clearly: what prior work exists, what gap remains, what this paper fills
- Draft a strong introduction that a non-expert reviewer can follow
- Give and receive constructive written peer review on a draft section

#### WEEKLY GOALS

- Literature review draft complete — organized by theme, building from seed papers outward to self-found sources
- Introduction draft from researcher — problem statement, gap, contribution, and paper organization
- All three participants have exchanged written peer review on at least one section
- Remaining citation gaps identified and assigned to close in week 9

#### STUDENTS

- Draft the literature review — four themes: (1) data poisoning in ML, (2) multi-object tracking and ReID, (3) multi-camera tracking and MTMC benchmarks, (4) adversarial attacks on trackers and detection methods
- Seed papers anchor each theme: Seed 1 → theme 1, Seed 2 → theme 2, Seeds 3 and 4 → themes 4
- Build outward from seeds to the papers found independently in weeks 1–7

#### RESEARCHER

- Draft the paper introduction: problem statement, why multi-camera embedding poisoning is a novel and serious threat, gap in prior work, and the paper's contribution
- Peer review both students' literature review drafts with written section-level feedback
- Ensure the contribution statement from week 7 is reflected consistently in the introduction

#### FRIDAY PRESENTATION

- Each student presents their literature review section draft — group gives feedback on thematic organization, gap coverage, and argument flow
- Researcher walks through introduction draft — group checks: is the contribution clear on first read?
- Each student presents 1–2 papers found to close remaining citation gaps

#### DELIVERABLES

- Literature review draft — thematically organized across four pillars, seeded by the four assigned papers
- Introduction draft from lead researcher — with clear problem, gap, and contribution statement
- Written peer review notes — each person gives section-level feedback on at least one other person's draft

### Week 9

PHASE 4  
WRITING &  
PUBLICATION

### Methods, Results & Contribution Framing

*Full rough draft assembled — all sections integrated into one document*

#### LEARNING OBJECTIVES

- Write a clear, reproducible methods section covering dataset, baseline, attack implementation, and detection method

#### WEEKLY GOALS

- Complete rough draft — all sections present and integrated into one document
- Contribution statement consistent from abstract through discussion

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| <ul style="list-style-type: none"><li>• Write a results section that tells a story: clean baseline → attack degradation → detection performance → scalability</li><li>• Ensure the contribution is stated consistently across introduction, methods, and discussion</li><li>• Read the full draft critically — as a reviewer, not as the author</li></ul>                                                                                                                                             |  | <ul style="list-style-type: none"><li>• Abstract finalized and accurately reflecting the full paper</li><li>• Every participant has flagged one thing that is unclear and one thing that is strong</li></ul>                                                                                                                                                                                          |  |
| <b>STUDENTS</b> <ul style="list-style-type: none"><li>→ Draft the methods section: dataset and scenario selection, BoT-SORT-ReID setup, embedding poison hook implementation (with pseudocode), cross-camera consistency detector, and evaluation protocol</li><li>→ Draft the results section: four subsections mirroring the four experiment types (baseline, attack, detection, scalability)</li><li>→ Integrate all sections into a single document with consistent notation throughout</li></ul> |  | <b>RESEARCHER</b> <ul style="list-style-type: none"><li>→ Synthesize all sections into a coherent full draft — check argument consistency throughout</li><li>→ Write or revise the abstract and conclusion</li><li>→ Ensure the threat model and contribution framing are consistent from introduction through discussion</li></ul>                                                                   |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  | <b>FRIDAY PRESENTATION</b> <ul style="list-style-type: none"><li>→ Full paper read-aloud walkthrough — read it as a first-time reviewer from abstract to conclusion</li><li>→ Each person flags one unclear element and one strong element</li><li>→ Final papers to fill remaining citation gaps presented</li><li>→ Group agrees on a revision plan for week 10 — who owns which sections</li></ul> |  |
| <b>DELIVERABLES</b> <ul style="list-style-type: none"><li>□ Complete rough draft — all sections present, integrated, and consistently notated</li><li>□ Abstract finalized by lead researcher</li><li>□ Revision plan for week 10 — documented, with section ownership assigned</li></ul>                                                                                                                                                                                                             |  |                                                                                                                                                                                                                                                                                                                                                                                                       |  |

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| <b>Week 10</b><br>PHASE 4<br>WRITING &<br>PUBLICATION                                                                                                                                                                                                                                                                                                                                                                                                 |  | <b>Revision, Venue Targeting &amp; Final Presentation</b><br><i>Paper polished, venue identified, final NDSU presentation delivered</i>                                                                                                                                                                                                                                                                                                                     |  |                                                                                                                                                                                                                                                                                                                            |  |
| <b>LEARNING OBJECTIVES</b> <ul style="list-style-type: none"> <li>• Revise the paper incorporating all group feedback from week 9</li> <li>• Identify target publication venues and understand their scope, format, and submission deadlines</li> <li>• Deliver a polished final presentation to NDSU covering the full research story</li> <li>• Establish a clear post-program submission path — who owns what after the experience ends</li> </ul> |  | <b>WEEKLY GOALS</b> <ul style="list-style-type: none"> <li>• Revised paper draft incorporating all feedback — publication-ready in structure and argument</li> <li>• 2–3 target venues identified with deadlines, format requirements, and page limits noted</li> <li>• Final NDSU presentation delivered — full story from problem to detection results</li> <li>• Post-program task assignments confirmed in writing by all three participants</li> </ul> |  |                                                                                                                                                                                                                                                                                                                            |  |
| <b>STUDENTS</b> <ul style="list-style-type: none"> <li>→ Revise assigned sections based on all group feedback from week 9</li> <li>→ Prepare and deliver the final NDSU presentation — tell the full story: why embedding poisoning is a real threat, what we built, what we found, and what comes next</li> </ul>                                                                                                                                    |  | <b>RESEARCHER</b> <ul style="list-style-type: none"> <li>→ Identify 2–3 target venues: CVPR adversarial ML workshop, IEEE S&amp;P main track or workshop, ECCV, USENIX Security, or ACM CCS depending on the paper's security vs. vision framing</li> <li>→ Assign post-program revision tasks — documented, with deadlines, so submission</li> </ul>                                                                                                       |  | <b>FRIDAY PRESENTATION</b> <ul style="list-style-type: none"> <li>→ Final presentation to NDSU — BoT-SORT-ReID pipeline, embedding-space poisoning attack, cross-camera consistency detector, scalability results</li> <li>→ Discuss publication next steps: venue shortlist, timeline, author responsibilities</li> </ul> |  |

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| → Write a personal research reflection: what would you pursue next, and what would you do differently?                                                                                                                                                                                                                                                                                                                                                                                                                                                | happens after the experience ends<br>→ Confirm each student's continued authorship role | → Each student shares one thing they are proud of from the experience<br>→ Researcher acknowledges each student's specific contribution to the paper and the science |
| <b>DELIVERABLES</b> <ul style="list-style-type: none"><li>□ Revised paper draft — all feedback incorporated, publication-ready structure</li><li>□ Final NDSU presentation deck</li><li>□ Publication venue shortlist — 2–3 venues with deadlines, format, and page limit requirements</li><li>□ Post-program task assignments — documented and confirmed by all three participants</li><li>□ Personal reflection note from each student</li></ul> <b>MILESTONE:</b> Program complete. Paper draft ready for submission. Post-program path confirmed. |                                                                                         |                                                                                                                                                                      |

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## Quick Reference — Recurring Formats

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### Annotated Bibliography Entry Format (6 fields — used every week)

| # | Field                | What to write                                                                               |
|---|----------------------|---------------------------------------------------------------------------------------------|
| 1 | Full citation        | Author(s), year, title, venue/journal, DOI or URL — IEEE or APA format                      |
| 2 | Argument summary     | 2–3 sentences: what does this paper claim and why does it matter?                           |
| 3 | Method used          | What data, experiments, or analysis did they use to support the claim?                      |
| 4 | Strength of evidence | How convincing is the evidence? What are the experimental conditions?                       |
| 5 | Limitation or gap    | What does the paper leave unanswered or assume away?                                        |
| 6 | Relevance to project | 1–2 sentences: how does this connect to embedding-space poisoning in multi-camera tracking? |

### Friday Paper Presentation Structure (10 minutes — same every week)

| # | Point                       | What to cover                                                                                |
|---|-----------------------------|----------------------------------------------------------------------------------------------|
| 1 | Paper ID                    | Title, authors, venue, year — 30 seconds                                                     |
| 2 | Main argument               | One sentence: what does this paper claim?                                                    |
| 3 | Method or evidence          | How did they show it? Dataset, experiment, proof, or analysis                                |
| 4 | Key finding or result       | The single most important thing the paper found or demonstrated                              |
| 5 | Limitation or open question | What does it leave unsolved? What are you skeptical of?                                      |
| 6 | Connection to project       | How does this change how you think about embedding-space poisoning in multi-camera tracking? |